

Discrete mathematics I

Complex numbers

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Spring 2026

Extension of number sets

- **Natural numbers:** $\mathbb{N} = \{0, 1, 2, \dots\}$

There is no natural number $x \in \mathbb{N}$ such that $x + 2 = 1$!

On $\mathbb{N}$ subtraction is not defined for all numbers.

- **Integers:** $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$

In $\mathbb{Z}$ subtraction is always possible: $x = -1$.

There is no integer $x \in \mathbb{Z}$ such that $x \cdot 2 = 1$!

On $\mathbb{Z}$ division is not defined by all numbers.

- **Rational numbers:** $\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}$

We can divide by any nonzero number in $\mathbb{Q}$: $x = \frac{1}{2}$.

There is no rational number $x \in \mathbb{Q}$ such that $x^2 = 2$!

Taking the square root of a rational number $\mathbb{Q}$ does not always produce a rational number, not even in the case of a nonnegative rational number.

- **Real numbers:** $\mathbb{R}$.

We can take the square root of any nonnegative number in $\mathbb{R}$.

There is no real number $x \in \mathbb{R}$ such that $x^2 = -1$!

We cannot take the square root of negative numbers in $x \in \mathbb{R}$, since:

$$\forall x \in \mathbb{R} : x^2 \geq 0.$$

Extension of number sets

Among **complex numbers** the equation $x^2 = -1$ can be solved!

Applications of complex numbers:

- solving equations;
- geometry;
- physics (fluid dynamics, quantum mechanics, relativity theory);
- computer graphics, quantum computers.

Introducing complex numbers

Definition (imaginary unit)

Let i be a solution to the equation $x^2 = -1$; i is called the **imaginary unit**.

We would like to extend the operations of addition and multiplication from the set of real numbers to a larger set containing i , while keeping the 'usual rules' of calculation and adding the rule: $i^2 = -1$. E.g.:

$$(1 + i)^2 = 1 + 2i + i^2 = 1 + 2i + (-1) = 2i$$

Definition of complex numbers (informal definition)

Definition (complex numbers)

The expressions of the form $a + bi$ where $a, b \in \mathbb{R}$, are called **complex numbers** with addition and multiplication defined as:

- **addition:** $(a + bi) + (c + di) = a + c + (b + d)i$.
- **multiplication:** $(a + bi)(c + di) = ac - bd + (ad + bc)i$.

The set of all complex numbers is denoted by $\mathbb{C}$. The form $a + bi$ where $a, b \in \mathbb{R}$ is called the **algebraic form** (or **Cartesian** or **rectangular form**) of a complex number.

Definition (real part and imaginary part of a complex number)

Let $z = a + bi$ ($a, b \in \mathbb{R}$) be a complex number. Then the **real part** of z is $\operatorname{Re}(z) = a \in \mathbb{R}$ and the **imaginary part** of z is $\operatorname{Im}(z) = b \in \mathbb{R}$.

- **Note:** $\operatorname{Im}(z) \neq bi$
- The complex numbers of the form $a + 0 \cdot i$ are the real numbers. The complex numbers of the form $0 + bi$ are called **pure imaginary numbers**.
- Two complex numbers with algebraic forms $a + bi$ and $c + di$ are equal: $a + bi = c + di$, if and only if $a = c$ and $b = d$.

The definition of complex numbers (formal definition)

Definition (formal definition of complex numbers)

The set $\mathbb{C}$ of **complex numbers** is the set $\mathbb{R} \times \mathbb{R}$ together with the following operations:

- **addition:** $(a, b) + (c, d) = (a + c, d + b)$;
- **multiplication:** $(a, b) \cdot (c, d) = (ac - bd, ad + bc)$.

The two definitions of complex numbers are equivalent: $a + bi \leftrightarrow (a, b)$,
e.g. $i \leftrightarrow (0, 1)$.

The format $a + bi$ is more convenient for manual calculations.

The format (a, b) is more convenient for use with computers.

Basic properties of addition and multiplication on $\mathbb{C}$

Based on the definitions it is easy to verify the following properties:

Proposition (Basic properties of operations on $\mathbb{C}$)

Properties of addition

- 1 Associativity: $\forall z_1, z_2, z_3 \in \mathbb{C} : (z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$.
- 2 Commutativity: $\forall z_1, z_2 \in \mathbb{C} : z_1 + z_2 = z_2 + z_1$.
- 3 Neutral element (zero element): $\exists 0 \in \mathbb{C}$ (*zero element*) such that

$$\forall z \in \mathbb{C} : 0 + z = z + 0 = z.$$

- 4 Additive inverse (opposite): $\forall z \in \mathbb{C} : \exists -z \in \mathbb{C}$ (*opposite of z*) such that

$$z + (-z) = (-z) + z = 0.$$

Properties of multiplication

- 1 Associativity: $\forall z_1, z_2, z_3 \in \mathbb{C} : (z_1 \cdot z_2) \cdot z_3 = z_1 \cdot (z_2 \cdot z_3)$.
- 2 Commutativity: $\forall z_1, z_2 \in \mathbb{C} : z_1 \cdot z_2 = z_2 \cdot z_1$.
- 3 Unit element: $\exists 1 \in \mathbb{C}$ (*unit element*) such that $\forall z \in \mathbb{C} : 1 \cdot z = z \cdot 1 = z$.
- 4 Multiplicative inverse (reciprocal): $\forall z \in \mathbb{C} \setminus \{0\} : \exists z^{-1} = \frac{1}{z} \in \mathbb{C}$ (*reciprocal of z*) such that $z \cdot z^{-1} = z^{-1} \cdot z = 1$.

Distributivity: $\forall z_1, z_2, z_3 \in \mathbb{C} : z_1(z_2 + z_3) = z_1z_2 + z_1z_3$
(and $(z_1 + z_2)z_3 = z_1z_3 + z_2z_3$)

$(\mathbb{C}, +, \cdot)$ is an algebraically closed field

Corollary:

- Because of the above properties, the algebraic structure $(\mathbb{C}, +, \cdot)$ is a so called *field* (just like $(\mathbb{R}, +, \cdot)$ and $(\mathbb{Q}, +, \cdot)$).
- Informally we can say that we can calculate with complex numbers 'in the same way' as with real numbers (in sums and products we can 'move' the brackets; the order of the terms in a sum and of the factors in a product can be changed; brackets can be expanded by the distributive property etc.) with the additional rule: $i^2 = -1$.

Fundamental Theorem of Algebra: It can also be shown – proof is not easy – that all polynomial equations of positive degree has solution in $\mathbb{C}$. Hence the field $(\mathbb{C}, +, \cdot)$ is *algebraically closed*. ('No need to introduce further numbers!')

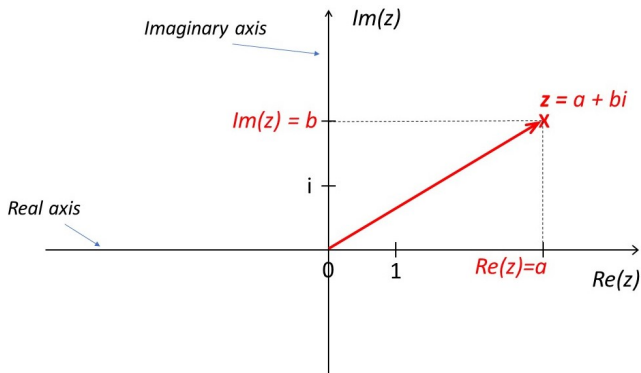
Theorem (Fundamental Theorem of Algebra; no proof required)

Let $n \in \mathbb{N}^+$. Then for every $a_0, \dots, a_n \in \mathbb{C}$, $a_n \neq 0$, there exists $z \in \mathbb{C}$ such that $a_0 + a_1 z + a_2 z^2 + \dots + a_n z^n = 0$ (i.e. the polynomial $a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$ has a root in $\mathbb{C}$.)

Representing complex numbers in the Complex plane (Gaussian plane, Argand diagram)

Complex numbers can be represented in the **complex plane** (**Gaussian plane, Argand diagram**):

- $z = a + bi \leftrightarrow (a, b)$
- bijection (one-to-one correspondence) between $\mathbb{C}$ and the points (or vectors) of the plane.



Calculating with complex numbers: absolute value, conjugate

Definition (absolute value of a complex number)

The **absolute value** of a complex number z with algebraic form $z = a + bi$ is $|z| = \sqrt{a^2 + b^2}$.

In particular, if z is a real number, then $z = a$ and its absolute value is the 'usual' absolute value of a real number: $|z| = |a| = \sqrt{a^2}$.

Proposition (Hw)

For any complex number z :

- 1 $|z| \geq 0$,
- 2 $|z| = 0 \Leftrightarrow z = 0$.

Definition (conjugate of a complex number)

The **conjugate** of a complex number z with algebraic form $z = a + bi$ is $\bar{z} = a - bi$.

Calculating with complex numbers: opposite, subtraction

Definition (opposite of a complex number)

The **opposite** of a complex number z is the complex number denoted by $-z$ such that $z + (-z) = 0$.

Proposition (Opposite of a complex number; proof is hw)

The opposite of a complex number z with algebraic form $z = a + bi$ is the complex number with algebraic form $-z = -a - bi$.

Definition (subtraction of complex numbers)

The **difference** of complex numbers z and w is defined as:

$$z - w = z + (-w)$$

Note: Equivalently, we could define the difference $z - w$ as the (unique) complex number with the property: $(z - w) + w = z$.

Calculating with complex numbers: reciprocal, division

Definition (reciprocal of a nonzero complex number)

The **reciprocal** of a nonzero complex number z is the number $z^{-1} = \frac{1}{z}$ such that $z \cdot z^{-1} = 1$.

Note: By the definition of multiplication it is easy to show that every nonzero complex number has a reciprocal, which is unique.

Using the reciprocal, we can define division by nonzero complex numbers:

Definition (division by nonzero complex numbers)

The **quotient** of two complex numbers z and $w \neq 0$ is:

$$\frac{z}{w} = z \cdot w^{-1}.$$

Note: Equivalently, we could define the quotient $\frac{z}{w}$ as the (unique) complex number with the property: $\frac{z}{w} \cdot w = z$.

Calculating with complex numbers: reciprocal, division

What is $\frac{2+3i}{1+i}$ in algebraic form?

Idea: Similar to the rationalization of the denominator in fractions of real numbers:

$$\begin{aligned}\frac{1}{1+\sqrt{2}} &= \frac{1}{1+\sqrt{2}} \cdot \frac{1-\sqrt{2}}{1-\sqrt{2}} = \frac{1-\sqrt{2}}{(1+\sqrt{2})(1-\sqrt{2})} = \frac{1-\sqrt{2}}{1^2 - \sqrt{2}^2} = \\ &= \frac{1-\sqrt{2}}{1-2} = -1 + \sqrt{2}\end{aligned}$$

Multiply both the numerator and the denominator **by the conjugate of the denominator**:

$$\frac{2+3i}{1+i} = \frac{2+3i}{1+i} \cdot \frac{1-i}{1-i} = \frac{(2+3i)(1-i)}{(1+i)(1-i)} = \frac{5+i}{1^2 - i^2} = \frac{5+i}{1-(-1)} = \frac{5+i}{2} = \frac{5}{2} + \frac{1}{2}i$$

Why did this method work? When multiplying the denominator $1+i$ by its conjugate $1-i$, the result (the new denominator) is a real number.

Calculating with complex numbers: reciprocal, division

Lemma

For any complex number z we have $z \cdot \bar{z} = |z|^2$ (hence $z \cdot \bar{z}$ is a real number).

Proof

Let $z = a + bi$ be the algebraic form of z . Then
 $z \cdot \bar{z} = (a + bi)(a - bi) = a^2 + b^2 = |z|^2$.

Hence:

Proposition (Calculating the quotient in algebraic form)

Let $z, w \in \mathbb{C}$, $w \neq 0$. Then the quotient $\frac{z}{w}$ in algebraic form can be found as:

$$\frac{z}{w} = \frac{z \cdot \bar{w}}{w \cdot \bar{w}}$$

Proof

Let $z = a + bi$ and $w = c + di$ ($a, b, c, d \in \mathbb{R}$). Then

$$\frac{z}{w} = \frac{z \cdot \bar{w}}{w \cdot \bar{w}} = \frac{(a+bi)(c-di)}{(c+di)(c-di)} = \frac{ac+bd+(bc-ad)i}{c^2+d^2} = \frac{ac+bd}{c^2+d^2} + \frac{bc-ad}{c^2+d^2}i.$$

Calculating with complex numbers

Theorem (Properties of conjugation and the absolute value of complex numbers; proof is hw.)

Let z and w be complex numbers. Then:

- 1 $\overline{\overline{z}} = z;$
- 2 $\overline{z + w} = \overline{z} + \overline{w};$
- 3 $\overline{z \cdot w} = \overline{z} \cdot \overline{w};$
- 4 $z + \overline{z} = 2\operatorname{Re}(z);$
- 5 $z - \overline{z} = 2\operatorname{Im}(z) \cdot i ;$
- 6 $z \cdot \overline{z} = |z|^2;$
- 7 if $z \neq 0$ then $z^{-1} = \frac{\overline{z}}{|z|^2};$
- 8 $|0| = 0$ and if $z \neq 0$ then $|z| > 0;$
- 9 $|\overline{z}| = |z|;$
- 10 $|z \cdot w| = |z| \cdot |w|;$
- 11 $|z + w| \leq |z| + |w|$ (triangle-inequality).

Calculating with complex numbers

Theorem

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$$\textcircled{10} \quad |z \cdot w| = |z| \cdot |w|;$$

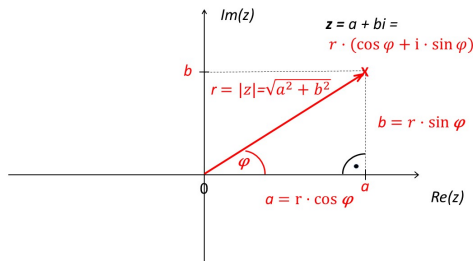
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Proof

$$|z \cdot w|^2 = z \cdot w \cdot \overline{z \cdot w} = z \cdot w \cdot \bar{z} \cdot \bar{w} = z \cdot \bar{z} \cdot w \cdot \bar{w} = |z|^2 \cdot |w|^2 = (|z| \cdot |w|)^2.$$

The polar form of complex numbers

Let $z = a + bi \in \mathbb{C}$ ($a, b \in \mathbb{R}$), $z \neq 0$.



- The length r of the vector (a, b) is: $r = \sqrt{a^2 + b^2} = |z|$.
- Denote by φ the angle from the positive real axis to the vector (a, b) (comment: this angle is not unique, because integer multiples of 2π can be added to it).

The coordinates a and b expressed in terms of r and φ ('polar coordinates'):

$$a = r \cdot \cos \varphi, \quad b = r \cdot \sin \varphi$$

The polar form of complex numbers

Definition (polar form)

The **polar form** of a nonzero complex number $z \in \mathbb{C}$ is:

$$z = r(\cos \varphi + i \sin \varphi)$$

where $r = |z|$.

Note:

- The polar form of zero is usually not used, because the angle could be any real number.
- The polar form is not unique (because the angle is not unique):
 $r(\cos \varphi + i \sin \varphi) = r(\cos(\varphi + 2\pi) + i \sin(\varphi + 2\pi)).$

Definition (argument)

The **argument** of a nonzero $z \in \mathbb{C}$ is the angle $\varphi = \arg(z) \in [0, 2\pi)$ such that $z = r(\cos \varphi + i \sin \varphi)$ where $r = |z|$.

Converting from algebraic form to polar form

Given the algebraic form $z = a + bi \neq 0$ we would like to determine the polar form of a nonzero complex number.

$$a + bi = r(\cos \varphi + i \sin \varphi)$$

Given a and b we are looking for $r = |z|$ and φ .

- Finding r : $r = |z| = \sqrt{a^2 + b^2}$.
- Finding φ : Since $a = r \cos \varphi$, hence

$$\varphi = \begin{cases} \arccos \frac{a}{r}, & \text{if } b \geq 0; \\ -\arccos \frac{a}{r}, & \text{if } b < 0. \end{cases}$$

Calculations in polar form

Theorem

Let $z, w \in \mathbb{C}$ be nonzero complex numbers: $z = |z|(\cos \varphi + i \sin \varphi)$, $w = |w|(\cos \psi + i \sin \psi)$, and let $n \in \mathbb{N}^+$. Then

- ① $zw = |z||w|(\cos(\varphi + \psi) + i \sin(\varphi + \psi))$ (Multiplication in polar form);
- ② $\frac{z}{w} = \frac{|z|}{|w|} \cdot (\cos(\varphi - \psi) + i \sin(\varphi - \psi))$ (Division in polar form);
- ③ $z^n = |z|^n(\cos n\varphi + i \sin n\varphi)$ (De Moivre's formula: raising complex numbers to the n^{th} power in polar form).

The angles are **added, subtracted, multiplied by n** .

Geometric meaning: Multiplication by a nonzero complex number $z \in \mathbb{C}$ acts on the complex plane like an **enlargement by a scale factor of $|z|$** together with a rotation by an angle of **$\arg(z)$** around the origin.

Proof

1

$$\begin{aligned}
 zw &= |z|(\cos \varphi + i \sin \varphi) \cdot |w|(\cos \psi + i \sin \psi) = \\
 &\quad \text{By commutativity of multiplication:} \\
 &= |z||w|(\cos \varphi + i \sin \varphi) \cdot (\cos \psi + i \sin \psi) = \\
 &\quad \text{By definition of multiplication:} \\
 &= |z||w|(\cos \varphi \cos \psi - \sin \varphi \sin \psi + i(\cos \varphi \sin \psi + \sin \varphi \cos \psi)) = \\
 &\quad \text{Hence by the trigonometric addition formulas:} \\
 &= |z||w|(\cos(\varphi + \psi) + i \sin(\varphi + \psi))
 \end{aligned}$$

Trigonometric addition formulas:

$$\cos(\varphi + \psi) = \cos \varphi \cos \psi - \sin \varphi \sin \psi$$

$$\sin(\varphi + \psi) = \cos \varphi \sin \psi + \sin \varphi \cos \psi$$

- The **absolute value** of the product: $|zw| = |z||w|$.
- The **argument** of the product:
 - if $0 \leq \arg(z) + \arg(w) < 2\pi$ then $\arg(zw) = \arg(z) + \arg(w)$;
 - if $2\pi \leq \arg(z) + \arg(w) < 4\pi$ then $\arg(zw) = \arg(z) + \arg(w) - 2\pi$.

The functions **sin**, **cos** are periodic with a period 2π , for finding the argument of the product, we may need to **reduce** the sum of the arguments by 2π .

Roots of complex numbers

Definition (n^{th} roots of a complex number)

Let $n \in \mathbb{N}^+$ and $z \in \mathbb{C}$. The n^{th} roots of z are those complex numbers w for which $w^n = z$.

Theorem (Formula for the n^{th} roots of a complex number)

Let $z = |z|(\cos \varphi + i \sin \varphi)$, $n \in \mathbb{N}^+$. The n^{th} roots of z are:

$$w_k = \sqrt[n]{|z|} \left(\cos \left(\frac{\varphi}{n} + \frac{2k\pi}{n} \right) + i \sin \left(\frac{\varphi}{n} + \frac{2k\pi}{n} \right) \right)$$

$$k = 0, 1, \dots, n-1.$$

The following fact will be used in the proof of the theorem:

Two complex numbers given in polar forms $z = |z|(\cos \varphi + i \sin \varphi)$ and $w = |w|(\cos \psi + i \sin \psi)$ are equal:

$$|z|(\cos \varphi + i \sin \varphi) = |w|(\cos \psi + i \sin \psi),$$

if and only if:

- $|z| = |w|$ and
- $\varphi = \psi + 2k\pi$ for some $k \in \mathbb{Z}$.

Roots of complex numbers

Theorem (Formula for the n^{th} roots of a complex number)

Let $z = |z|(\cos \varphi + i \sin \varphi)$, $n \in \mathbb{N}^+$. The n^{th} roots of z are:

$$w_k = \sqrt[n]{|z|} \left(\cos \left(\frac{\varphi}{n} + \frac{2k\pi}{n} \right) + i \sin \left(\frac{\varphi}{n} + \frac{2k\pi}{n} \right) \right)$$

$$k = 0, 1, \dots, n-1.$$

Proof

By De Moivre's formula, for any complex number $w = |w|(\cos \psi + i \sin \psi)$ we have $w^n = |w|^n(\cos n\psi + i \sin n\psi)$.

Hence $w^n = z$ is equivalent to $|w|^n(\cos n\psi + i \sin n\psi) = |z|(\cos \varphi + i \sin \varphi)$, which holds if and only if:

- $|w|^n = |z| \Leftrightarrow |w| = \sqrt[n]{|z|}$ and
- $n\psi = \varphi + 2k\pi$ for some $k \in \mathbb{Z} \Leftrightarrow \psi = \frac{\varphi}{n} + \frac{2k\pi}{n}$ for some $k \in \mathbb{Z}$.

The values $k \in \{0, 1, \dots, n-1\}$ all yield distinct n^{th} roots, and no further numbers are obtained when k is outside this range.

Example

Example

Find the 6th roots (w) of $\frac{1-i}{\sqrt{3}+i}$.

$$1 - i = \sqrt{2} \left(\frac{\sqrt{2}}{2} - i \frac{\sqrt{2}}{2} \right) = \sqrt{2} \left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4} \right)$$

$$\sqrt{3} + i = 2 \left(\frac{\sqrt{3}}{2} + i \frac{1}{2} \right) = 2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$$

$$\text{Since } \frac{7\pi}{4} - \frac{\pi}{6} = \frac{19\pi}{12}, \text{ hence: } \frac{1-i}{\sqrt{3}+i} = \frac{1}{\sqrt{2}} \left(\cos \frac{19\pi}{12} + i \sin \frac{19\pi}{12} \right).$$

So the 6th roots are:

$$w_k = \frac{1}{\sqrt[12]{2}} \left(\cos \frac{19\pi+24k\pi}{72} + i \sin \frac{19\pi+24k\pi}{72} \right) : k = 0, 1, \dots, 5$$

Complex roots of unity

Definition (n^{th} roots of unit)

For any $n \in \mathbb{N}^+$ the n^{th} roots of 1 are called the n^{th} roots of unity. (i.e. the complex numbers ε satisfying $\varepsilon^n = 1$.)

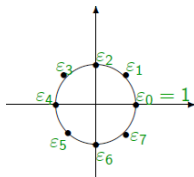
Using the formula of the n^{th} roots of complex numbers we obtain the following:

Theorem (The polar form of the n^{th} roots of unity)

For any $n \in \mathbb{N}^+$ the n^{th} roots of unity are:

$$\varepsilon_k = \varepsilon_k^{(n)} = \left(\cos \frac{2k\pi}{n} + i \sin \frac{2k\pi}{n} \right) : k = 0, 1, \dots, n-1.$$

Example: The 8^{th} roots of unity:



Roots of complex numbers and roots of unity

Theorem (Expressing all n^{th} roots of a complex number using one n^{th} root and the n^{th} roots of unity)

Let $z \in \mathbb{C}$ be a nonzero complex number, $n \in \mathbb{N}^+$ and $w \in \mathbb{C}$ be such that $w^n = z$. Then the n^{th} roots of z can be expressed in the following form:

$$w_k = w \varepsilon_k^{(n)} \text{ where } k = 0, 1, \dots, n-1.$$

Proof

All numbers of the form $w \varepsilon_k$ are n^{th} roots of z :

$(w \varepsilon_k)^n = w^n \varepsilon_k^n = z \cdot 1 = z$. These are n distinct values, hence we have obtained all n^{th} roots of z .

Primitive n^{th} roots of unity

Definition (roots of unity)

A complex number w is called a **root of unity** if it is an n^{th} root of unity for some $n \in \mathbb{N}^+$, i.e. if $w^n = 1$ for some $n \in \mathbb{N}^+$.

Definition (order of a root of unity, primitive n^{th} roots of unity)

The **order** of a root of unity w , denoted by $o(w)$, is the smallest $n \in \mathbb{N}^+$ such that $w^n = 1$. If the order of a root of unity is equal to n , then we call it a **primitive n^{th} root of unity**.

The order of an n^{th} root of unity is not necessarily equal to n :

Example

The 4^{th} roots of unity are $1, i, -1, -i$.

- $1, 1, 1, \dots \Rightarrow o(1) = 1$. The order of 1 is 1 .
- $-1, 1, -1, 1, \dots \Rightarrow o(-1) = 2$. The order of -1 is 2 .
- $i, -1, -i, 1, i, -1, \dots \Rightarrow o(i) = 4$. The order of i is 4 .
- $-i, -1, i, 1, -i, -1, \dots \Rightarrow o(-i) = 4$. The order of $-i$ is 4 .

Therefore the **primitive 4^{th} roots of unity** are i and $-i$.

Primitive n^{th} roots of unity: generating all n^{th} roots of unity

Proposition (Generating all n^{th} roots of unity by a primitive n^{th} root of unity)

If ε is a primitive n^{th} root of unity, then the list $\varepsilon^0, \varepsilon^1, \dots, \varepsilon^{n-1}$ is a list of all n^{th} roots of unity.

Proof

Since ε is an n^{th} root of unity, hence for every $k \in \mathbb{N}$:
 $(\varepsilon^k)^n = \varepsilon^{kn} = (\varepsilon^n)^k = 1^k = 1$. Therefore $\varepsilon^0, \varepsilon^1, \dots, \varepsilon^{n-1}$ are all n^{th} roots of unity. For any $0 \leq i < j \leq n-1$: $j-i < n = o(\varepsilon)$, and so $\frac{\varepsilon^j}{\varepsilon^i} = \varepsilon^{j-i} \neq 1$, hence, $\varepsilon^i \neq \varepsilon^j$. Therefore $\varepsilon^0, \varepsilon^1, \dots, \varepsilon^{n-1}$ are n distinct numbers, and so this list contains all n^{th} roots of unity.

Primitive n^{th} roots of unity: polar form

Example

- Primitive 1. root of unity: 1 ;
- Primitive 2. roots of unity: -1 ;
- Primitive 3. roots of unity: $\frac{-1 \pm i\sqrt{3}}{2}$;
- Primitive 4. roots of unity: $\pm i$;
- Primitive 5. roots of unity: $\dots$ (HW)
- Primitive 6. roots of unity: $\frac{1 \pm i\sqrt{3}}{2}$.

Proposition (Polar forms of the primitive n^{th} roots of unity; no proof required)

An n^{th} root of unity $\cos\left(\frac{2k\pi}{n}\right) + i \sin\left(\frac{2k\pi}{n}\right)$ is a primitive n^{th} root of unity if and only if $\gcd(n, k) = 1$.